

I am a robotics researcher with a focus on building intelligent, generalizable autonomous agents that operate in complex physical environments. I explore the intersection of **Robotics, Generative Modeling, Computer Vision and Reinforcement learning**, with a particular emphasis on learning from limited supervision and unstructured data. My PhD work aims to bridge perception, prediction, and control by developing compositional and causal ML models that enable robust decision-making and planning in real-world scenarios. I have a proven track record of building production-scale ML systems that integrate perception, prediction, and control for autonomous agents. My contributions span behavior prediction, multi-agent forecasting, sensor fusion, and efficient world models, with applications across autonomous systems, real-time control, and adaptive planning. Earlier in my career, I led a 20-member student autonomous vehicle team, deploying complete perception, planning, and control stacks on real hardware. These experiences reflect my commitment to bridging rigorous research with real-world impact—closing the loop between simulation, foundation models, and physical robotics.

🔑 I am currently seeking **full-time opportunities starting October 2025**.

✉ Reach me at: [hkumawat6\[at\]gatech.edu](mailto:hkumawat6@gatech.edu)

-->